

Progress Test 2.B — Thermo-S1

2.B · Thermoelastic behaviour of gases

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

A gas has a piezothermic coefficient $\beta = 3.66 \times 10^{-3} \text{ K}^{-1}$. Express β in $^{\circ}\text{C}^{-1}$.

A temperature *difference* of 1 K equals a difference of 1 $^{\circ}\text{C}$, so
 $\beta = 3.66 \times 10^{-3} \text{ }^{\circ}\text{C}^{-1}$ (same value).

Marking

①
0 1

Marking scheme

① 1 pt $3.66 \times 10^{-3} \text{ }^{\circ}\text{C}^{-1}$ (no +273.15 here).

Examiner's tip

A coefficient “per kelvin” is defined through a derivative, i.e. a temperature *difference*. Adding 273.15 would be a classic mistake.

Exercise 2. Piezothermic coefficient — definition

(5 marks)

Define the piezothermic coefficient β of a gas. Give its formula, the physical variation it measures, its sign and its SI unit. How does it differ from α ?

$\beta = \frac{1}{p} \left(\frac{\partial p}{\partial T} \right)_V$

It measures the **relative increase of the pressure per kelvin at constant volume**, for example a gas heated in a rigid, sealed container. Heating increases the pressure, so $\beta > 0$.

Unit: K^{-1} . By contrast, $\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_p$ describes the change of *volume* at constant *pressure*.

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① **2 pts** $\beta = \frac{1}{p} \left(\frac{\partial p}{\partial T} \right)_V$, with V fixed.
② **2 pts** Meaning: change of p with T at constant V (1 pt); $\beta > 0$ (1 pt).
③ **1 pt** K^{-1} and the difference with α .

Exercise 3. Gases at constant volume

(3 marks)

A gas enclosed in a rigid, sealed container is heated. Its pressure:

☒ increases ☐ decreases ☐ does not change ☐ becomes zero

Why: $\left(\frac{\partial p}{\partial T} \right)_V > 0$, i.e. $\beta > 0$ (equation (10) of the notes).

The coefficient β measures the variation of:

☐ V at fixed p ☒ p at fixed V
☐ T at fixed V ☐ V at fixed T

Why: see Tab. 3 of the notes. α : V at fixed p ; χ_T : V at fixed T .

For a car tyre (airtight, rigid rim), the variables held fixed are:

☒ n and V ☐ n and p ☐ T and V ☐ p and V

Why: Tab. 5: the tyre is airtight (n fixed) and the rim is rigid (V fixed). T and p rise because of friction.

Exercise 4. Tyre pressure

(5 marks)

A car tyre, whose volume is fixed, is inflated to p_1 at T_1 . While driving, its temperature rises by ΔT . Using the value of β given below (assumed constant), estimate the pressure increase Δp and the final pressure p_2 .

$p_1 = 2.0 \text{ bar}$	$T_1 = 300 \text{ K}$	$\Delta T = 30 \text{ K}$	$\beta = \frac{1}{300} \text{ K}^{-1}$
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At fixed V : $dp = \beta p dT$, so for a small change $\Delta p \approx \beta p_1 \Delta T$.

$p_1 = 2.0 \text{ bar} = 2.0 \times 10^5 \text{ Pa}$ $\Delta p \approx \frac{1}{300} \times 2.0 \times 10^5 \times 30$

$\Delta p \approx 2.0 \times 10^4 \text{ Pa} = 0.20 \text{ bar}$

$p_2 \approx 2.2 \text{ bar}$: a rise of 10 %, the same as $\Delta T/T_1$. (For such a gas $\beta = 1/T$, see 3.A, and p/T stays constant: $p_2 = p_1 T_2/T_1$ exactly.)

Marking

①
0 1
☐ ☐
②
0 1
☐ ☐
③
0 1
☐ ☐
④
0 1
☐ ☐
⑤
0 1
☐ ☐

Marking scheme

- ① 1 pt $\Delta p \approx \beta p \Delta T$ from the definition at constant V .
- ② 1 pt $p_1 = 2.0 \times 10^5 \text{ Pa}$.
- ③ 1 pt $\Delta p \approx 2.0 \times 10^4 \text{ Pa}$.
- ④ 1 pt $\Delta p = 0.20 \text{ bar}$ and $p_2 \approx 2.2 \text{ bar}$.
- ⑤ 1 pt Relative increase 10 % = $\Delta T/T_1$ noted.

Examiner's tip

Tyre pressures must be checked *cold*: a hot tyre reads about 10 % higher.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times$ 20/14 = / 20